

TITLE OF THE PROJECT

Accident Prevention in Hilly Areas Using IR Sensors

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ABSTRACT.

Road accidents in hilly areas are a major concern due to sharp curves, narrow roads, poor visibility, fog, and steep slopes. Many accidents occur because drivers are unaware of vehicles approaching from the opposite direction at blind turns. This project proposes an accident prevention system for hilly areas using IR sensors and a microcontroller to enhance road safety.

The system uses Infrared (IR) sensors installed on both sides of a curved or blind road to detect the presence of vehicles. When a vehicle is detected by the IR sensor on one side, a signal is sent to the Arduino-based control unit, which activates warning indicators such as LED traffic lights and a buzzer on the opposite side of the curve. This alerts approaching drivers to slow down and drive cautiously. Once the vehicle passes, the system automatically turns OFF the warning indicators.

This low-cost and energy-efficient system helps reduce head-on collisions and improves driver awareness in dangerous hilly regions. The proposed model is simple, reliable, and suitable for real-time implementation on mountain roads, hairpin bends, and narrow pathways.

AIM & OBJECTIVES

AIM:-To enhance road safety in hilly areas by detecting vehicles at blind turns and alerting drivers using IR sensors.

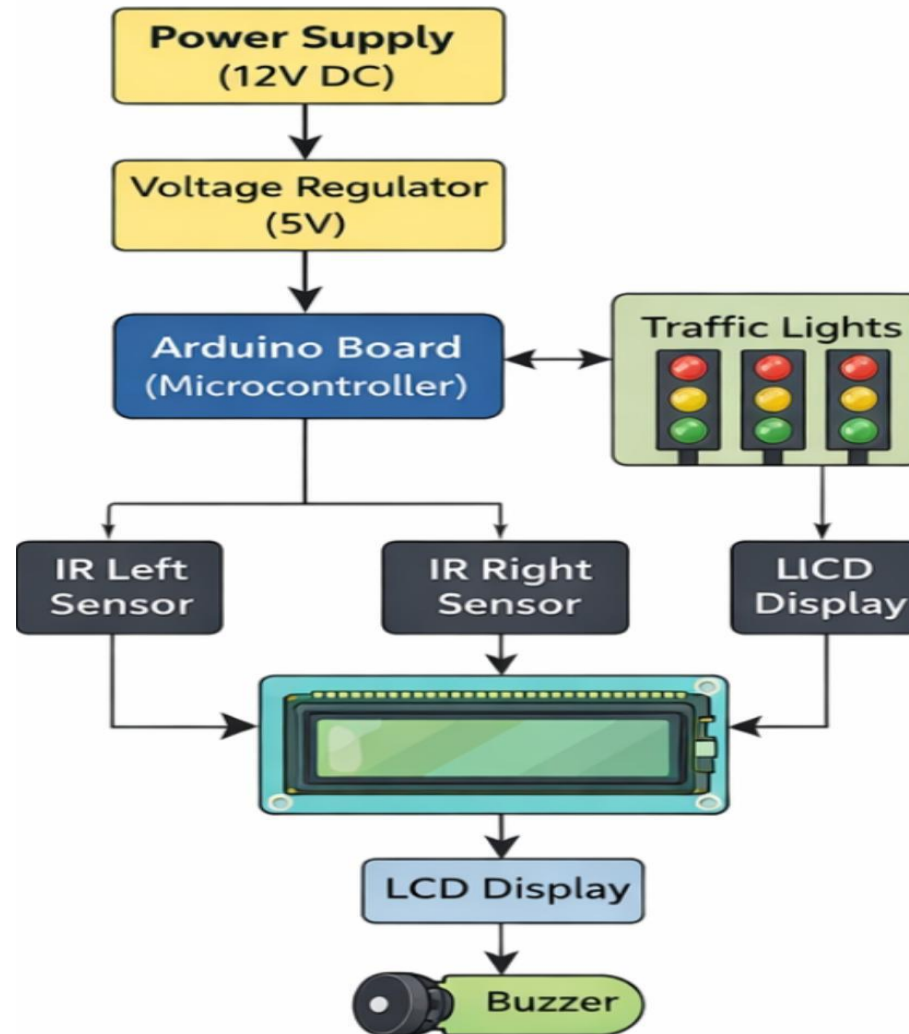
OBJECTIVES:-

- To detect the presence of vehicles approaching blind curves in hilly areas using IR sensors.
- To provide early warning signals to drivers coming from the opposite direction to reduce collision risk.
- To design an automatic system that operates without human intervention
- To improve road safety in accident-prone hilly regions with poor visibility.
- To develop a low-cost, reliable, and energy-efficient accident prevention solution.
- To minimize head-on collisions at narrow roads and hairpin bends.
- To implement a real-time alert mechanism using visual (LED) and optional audio (buzzer) indicators.
- To ensure easy installation and maintenance of the system in remote hilly areas.

EXISTING METHODOLOGY

- In the existing system, accident prevention in hilly areas mainly depends on **manual and passive safety methods**. These include **warning signboards, convex mirrors, speed limit boards, and road markings** placed near blind curves and narrow roads. Drivers rely on their own judgment and visibility to avoid collisions. In some locations, **human traffic control** is used during peak hours.
- However, these methods have several limitations. Warning boards and mirrors are **not effective during fog, rain, nighttime, or poor lighting conditions**. Convex mirrors can become unclear due to dust, rain, or damage. Human monitoring is **not available 24/7** and is impractical in remote hilly regions. Moreover, the existing system **does not provide real-time information** about approaching vehicles, increasing the risk of head-on accidents at blind turns.
- As a result, many accidents still occur due to delayed driver response and lack of automatic warning mechanisms, highlighting the need for an **automated accident prevention system**.

BLOCK DIAGRAM



PROPOSED METHODOLOGY

- The proposed methodology involves installing IR sensors on both sides of blind curves in hilly areas to detect approaching vehicles. When a vehicle is detected, the IR sensor sends a signal to the Arduino-based control unit, which processes the information and activates warning indicators such as LEDs, a buzzer, or caution signboards on the opposite side of the curve to alert drivers in advance.
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- Once the vehicle passes the curve, the IR sensor no longer detects its presence, and the control unit automatically turns OFF the warning indicators. This system operates continuously in real time with a stable power supply, providing a low-cost, reliable, and automatic solution for accident prevention in hilly regions.

REFERENCES

- Hilly Area Accident Prevention System Based on Infrared Sensors” Proposes a system using IR & PIR sensors with GPS, LED indicators & buzzer for mountainous roads.
- **VLC-based Collision Avoidance System for Vehicles on Curved Roads in Hilly Areas** — *Journal of Optical Communications (2025)*.
- **Smart Vehicle Accident Prevention and Road Safety System with Real Time Data Acquisition** — *International Journal of Engineering and Manufacturing (2024)*.

Thank you!